**Comparing two supermarket layouts: the effect of a middle aisle on basket size, spend, trip duration and endcap use**

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## Abstract

This paper aims to empirically establish the effectiveness of a supermarket layout with a middle aisle splitting all other aisles, compared to a 'traditional' layout (without a middle aisle). Two supermarkets in Australia were matched by size, turnover, population/location, and general layout, except that Store 2 had a middle aisle. The research aims to: (1) explore the shopper traffic entering and existing the middle aisle, and interaction with endcap promotions (which have effectively been doubled by the creation of the middle aisle in Store 2), and (2) compare the two stores based on basket size (in items and dollars) and trip duration. These are typical supermarket metrics, which might differ due to increased opportunities for impulse purchases from the extra endcap displays in the middle aisle. Despite the middle aisle, results show that all performance metrics are almost identical between the two stores on the overall level. However, the store containing the middle aisle sees a higher proportion of trips for under 15 items. Overall use of promotional ends of aisles is the same across both stores, meaning that the use of each endcap is effectively halved.

This means that the presence of a middle aisle does not bring any additional value in terms of making the store easier or quicker to navigate. It may provide additional benefits to retailers in terms of giving more promotional space to manufacturers, at the expense of making each endcap half as likely to generate interest. Our additional analysis of the traffic flow near the endcaps suggests that, in most occasions, shoppers pass through the aisle as if there was no break. These results are useful insights to retailers considering remodeling an existing store and manufactures buying endcap space in a store with a middle aisle.

*Keywords: supermarket layout; store layout; basket size; trip duration*

## Introduction and Background

### Physical retailing in context

Retailing is undergoing a transformation towards digitalization (Hagberg et al., 2016), though progress is slow. The focus in both the academic literature and trade press is increasingly directed towards online retail and the changing retail landscape due to technological advancements, but the vast majority of shopping is still done in brick and mortar stores: less than 10% by value is done online (U.S. Census Bureau News, 2018). Consumers prefer to shop in physical stores

(Smith and Anderson, 2016), and indeed the largest online retailer, Amazon, is experimenting with physical stores for books and groceries with new layouts, not requiring physical cash registers (Bensinger and Morris, 2014). While shoppers intermix and merge the digital and physical worlds (Licoppe, 2013), it is the physical design and layout of a supermarket which plays a major role in how easy or otherwise it is for consumers to complete shopping trips in a timely and cost-effective manner (Sorensen, 2016). Given the continued importance of physical stores, and the potential for innovation in this space, we investigate one particular innovation: a transverse aisle in a supermarket, splitting a traditional store in two.

Research in this space can be classified into three areas: rule-based data mining, online store layouts, and methods for assortment management (Altuntas, 2017). While assortment management is an established area of research which demonstrates convincingly that shelf space allocation has effects on sales (e.g. Desmet and Renaudin, 1998) and customer perceptions (Pizzi and Scarpi, 2016), overall retail store layout is a relatively new area, with challenges not met by the existing manufacturing and warehousing layout literature (Mowrey et al., 2017). Some initial work has begun in this field (see Yapicioglu and Smith, 2012), and the consideration of the shopper as an independent agent has led to the investigation of optimum shelf orientations for exposure (Parikh and Mowrey, 2014), which leads to suggestions for layouts optimized for viewability (Mowrey et al., 2017). The current research brings a new perspective of simple, documented, structured observation in the Ehrenbergian tradition (see Ehrenberg, 1969, 1995), in order to build theory from repeated observations. It is important to note that despite the positivist epistemological approach this implies, this research does not subscribe to rationalist models of human behaviour, but rather holds the view that consumers are responding to the competing mental and physical availability of brands and products (Sharp, 2010), and that changes to the layout of a store primarily influence the physical availability of products, which may have influences on their perceived or actual choices.

**The role of store layout**

The traditional layout of the main part of a supermarket comprises many parallel aisles, typically about 1.8-2 meters wide, and stretching for up to 30 meters long. Some authors have criticized these layouts, referring to particularly tall aisles as "canyons", where shoppers might feel "trapped" (Sorensen, 2009), and that the layout is not the optimum merely because it has existed for some time, and become taken for granted (Juel-Jacobsen, 2015b). The concept of the 'butt brush' law also points to the uncomfortable feeling that shoppers might experience in narrow aisles from being too close to each other, or store fixtures (Underhill, 1999). The store layout could affect many store performance metrics: how long it takes to shop a store (with the understanding that time is as important currency as money (Sorensen, 2009)), how easy it is to complete all tasks, and find needed items (Hui et al., 2012), which could influence basket size in terms of items and dollars.

In a standard store, shoppers circulate around the perimeter of the store (the "racetrack"), are more frequently seen at the ends of aisles rather than the middle, and display a "U-turn" pattern, where shoppers use the first few aisles to return to the front of the store (Caruso et al., 2017; Farley and Ring, 1966; Sorensen, 2016).

An alternative supermarket design has emerged: a supermarket with a middle aisle, which effectively splits the store in half, by crossing all the 'traditional' aisles at a 90 degree angle (see Figure 1). The intent of this layout is to 'break' the long aisles and offer shoppers a 'way out' or an opportunity to change direction in their navigation. This might change the patterns of behaviour seen above by giving shoppers alternative pathways between aisles. Rather than having no choice but to use the perimeter of the store, shoppers can now move between aisles more freely. This might result in a faster, more efficient and hence pleasing shopping experience, or the increased complexity of the environment may result in a more cognitively demanding, and so less efficient and pleasing shopping experience.

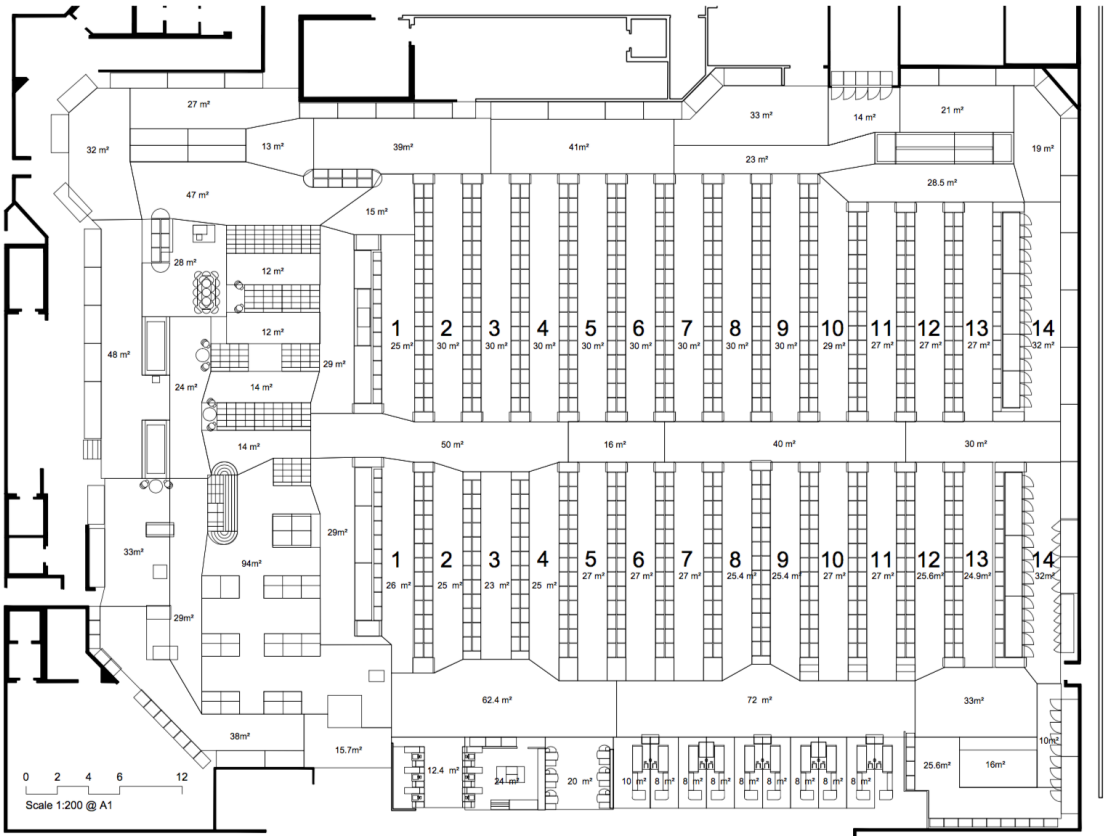



*Figure 1: A supermarket with a "split aisle". Aisles 1-7 are considered the "first half" of the store.*

While no research has specifically evaluated the effectiveness of supermarkets laid out with a middle aisle, such analysis could inform and justify potential costs for retailers to remodel an existing store, or inform the design of new stores, beyond the interrelationship between architecture, urban planning, and store design (Juel-Jacobsen, 2015a). Existing studies of the physical layout of stores

show that differences in arrangements of product offerings can influence product sales by -2% to +11% (Bezawada et al., 2009; Larson, 2006).

## Endcap displays

A potential benefit of the middle aisle for retailers and manufacturers is that it effectively doubles the amount of end-of-aisle (also called endcap) promotional spaces. Endcaps have been shown to be the most effective promotional spaces in a supermarket (Chevalier, 1975; Nakamura et al., 2014; Tan et al., 2018; Wilkinson et al., 1982). In these stores, all endcaps contained items which could be purchased at a discount. At the front of the store, the endcaps also contained cardboard promotional displays.

The main reason for endcaps effectiveness is their prominent location in the areas of the higher shopper traffic (along the race-track (Sorensen, 2016)). Higher traffic converts in higher visual attention (Caruso et al., 2017), and, eventually, higher sales than promotions in other parts of a store (Tan et al., 2018). Whether endcaps located in the middle aisle would enjoy the same levels of high traffic and associated uplift in sales, is yet unknown and will be explored in this article.

## Supermarket performance metrics

While the "effectiveness" of supermarkets can be assessed from multiple perspectives, including the actual shopper satisfaction with the store, satisfaction of the manufacturer, the retailer themselves are the ones who hold the power to make decisions about layout (Bogomolova et al., 2017). While other metrics such as customer satisfaction or the visual appeal of the layout are important considerations, retail businesses are largely focused on maximizing revenue per square meter of a store (Bogomolova et al., 2017). With this focus in mind, the ultimate success of a supermarket layout should be evaluated against typical store-level performance metrics. Past studies (Sorensen, 2016; Sorensen et al., 2017) have suggested that basket size (in terms of the number of items, and in dollar value), trip duration and traffic flow are indicative of a supermarket performance.

Basket size can be measured in two ways: either by the total number of items in a basket, or the total unique number of stock-keeping units (henceforth SKU). Each individual barcode in a store is considered an SKU. The number of unique SKUs measured is preferable in this context as it is a better measure of shopper navigation behaviour and removes the influence of quantity purchased from the measure. As an example, we would not want information about a shopper deciding to purchase the small, medium, or large pack, and similarly we do not need to know if a shopper decides to buy two of the same item. Basket sizes follow a mixed Poisson lognormal distribution (Sorensen et al., 2017), which is very heavily skewed to small basket sizes. In practice, this means that while practitioners often use average basket size, median is a more meaningful measure.

Trip duration is one of the resources customers spend while in store, in addition to money and angst (Sorensen, 2016). Shoppers have limited quantities of each resource, and once they have depleted any of the three, will leave the store (Sorensen, 2016). If a shopper depletes their time or angst reserves before depleting their fiscal budget, then a store is not maximizing profitability. A middle aisle may make the store easier and faster for shopper to navigate. A store which is faster for shoppers to navigate may not only reduce use of their time, but also of their angst resource, resulting in overall more satisfied and profitable shoppers.

**Research questions**

This article addresses the following research questions:

RQ1: How do the two stores (with and without a middle aisle) differ across typical supermarkets performance metrics:

- RQ1a: the number of items in a basket;
- RQ1b: dollars spent per trip;
- RQ1c: minutes spent per shopping trip (trip duration);

RQ2: How do shoppers traverse and interact with endcap promotions in the middle aisle (of Store 2)?

These insights would provide solid empirical evidence to retailers regarding potential benefits of adopting a middle aisle supermarket layout, helping them to make better-informed store layout decisions. Our insights from endcap traffic analysis in the middle and back of store aisles would also inform manufacturers regarding the comparative value of certain endcap locations, informing their negotiations with retailers regarding which endcaps to request for in-store promotion campaigns.

Our secondary aim is to investigate the traffic flow and shopper interactions with the endcaps in both stores, to determine the value that a supermarket might achieve by having a middle aisle and potentially doubling the endcap shelf space. A doubling of endcap (promotional) space could lead to more impulse purchases.

## Method

**Approach**

To investigate the effects of layout, laboratory research is often used, but the external validity of research not set in-store is limited (Smith and Burns, 1996). The study of supermarket layouts (e.g.Anic et al., 2010; Farley and Ring, 1966) often focuses on what *is* rather than what *could be*, and so few examples of research where layouts are compared exist. Smith et al. (1996) compare the differences in shoppers' *perception* of a store based on various ways of stacking a "power aisle", and Sommer and Aitkens (1982) compared two stores on the basis of the ability of shoppers to recall the locations of products. Both approaches rely on shoppers' abilities to notice supermarket environment in details and later recall it – abilities which have been shown to be very limited in routine

supermarket shopping, often described as being done on auto-pilot (Scamell-Katz, 2012; Sorensen, 2016). The limited ability of shoppers' to note and recall a supermarket environment questions the reliability of shopper-reported data, suggesting more reliable data such as real sales and non-intrusive observations would yield a more accurate account of real shopper behaviour in response to a supermarket environment. Further, direct comparison of two real-life stores is the best way to determine the effects of different supermarket layout (superior to any other lab or simulated environments).

**Store selection**

In this study, we compare two supermarkets of the same Australian chain, which were carefully selected to match by size, turnover, location, and demographics of the surrounding areas, and the store layout, except for one main difference - Store 2 had a middle aisle. The two stores selected had weekly turnover of $1.1-$1.2m, and were located in the suburbs of a major Australian capital city.

**Metrics**

To address RQ1, traffic flow was observed around the endcap promotion areas – that is the back of both stores (the racetrack path), and the middle aisle in Store 2. Audits of each store were made by dividing the floorplan up into "zones" whereby researchers could mark the location of each shopper they saw, and the direction they were facing, on trips through the store made at ten-minute intervals. Shoppers were considered to be interacting with an endcap if they were seen to have stopped walking in front of the display, but did not have to physically reach out to touch a product; merely looking was enough. In the standard store, 8624 shoppers were observed in situ, and in the mid-aisle store, 7610 shoppers were observed, over twelve hours per store.

Shopper behaviour in the middle aisle was observed from three perspectives: entering the aisle, exiting the aisle, and the use of promotional displays in the aisle. After completing a whole-store audit, researchers waited at three designated locations for 45 seconds each, selected so as to view the entire aisle without obstruction, and with the ability to bound the edges of the observed area easily. Data was recorded using a map of the aisle with spaces to record the direction of travel they saw shoppers walking, and which endcaps they saw people interacting with. For more details on this method, please see Page (2017).

To address RQ1a-b, basket size (in number of items and dollars spent) information was computed directly from retailer transaction logs. For RQ1c, shopper trip duration was collected using the Bluetooth method described and validated by Phua et al (2015). Specifically, the Bluetooth MAC addresses of mobile phones carried by shoppers were logged at the entrance/exit to the stores, and the trip time calculated by comparing the first and last signals received.

**Analysis**

The comparison of the above metrics across the two stores follows the same analysis procedure as demonstrated in Sorensen et al (Sorensen et al.), by

comparing central tendencies, as well as the distributions of the key metrics to account for the highly skewed, non-normal distributions of all metrics.

First, we compare the observed distributions of items bought, dollar spend, and trip duration across the two stores from both datasets to see if they are different. We then fit the observed distributions of each store to the benchmark Poisson lognormal and lognormal models proposed by Sorensen et al. (2017) to see if there is any deviation from the fundamental patterns in retailing formats.

Sorensen et al. (2017) show that trip duration fits the lognormal model. This model is not only empirically grounded but also appeals theoretically (see Sorensen et al., 2017) The probability function of the model is as per below with parameters $\mu$ and $\sigma$. We therefore use this model to benchmark the distribution of trip duration in both stores. We also extend Sorensen et al. (2017) by using this model to benchmark dollar spend.

$$f(\lambda;\mu,\sigma) = \frac{1}{\lambda\sigma\sqrt{2\pi}}\exp\left\{-\frac{(\log\lambda-\mu)^2}{2\sigma^2}\right\} \quad (1)$$

where, $\mu$ is the mean and $\sigma$ is the standard deviation of the normal distribution

$Y$ where $Y = \log(\lambda)$.

Similar to Sorensen et al. (2017) we use the Poisson lognormal to model the number of items bought. The probability function of the model as below with also two parameters $\mu$ and $\sigma$

$$\begin{aligned} f(x)_{PLN} &= \int_0^\infty f(x)_{poisson} f(\lambda;\mu,\sigma)d\lambda \\ &= \frac{1}{x!\sigma\sqrt{2\pi}}\int_0^\infty \lambda^{x-1}\exp(-\lambda)\exp\left\{-\frac{(\log\lambda-\mu)^2}{2\sigma^2}\right\}d\lambda \end{aligned} \quad (2)$$

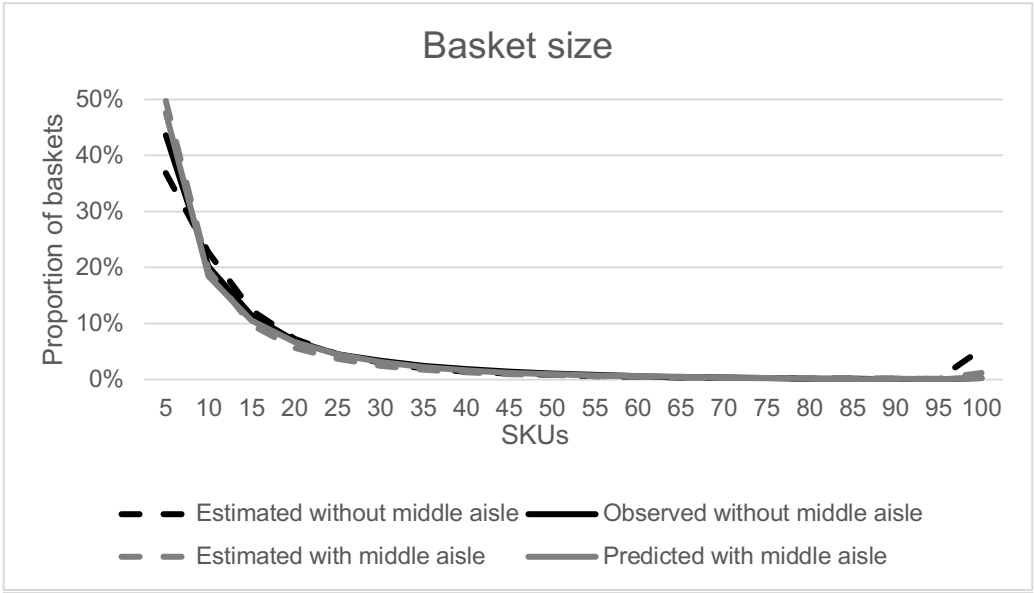
## Results

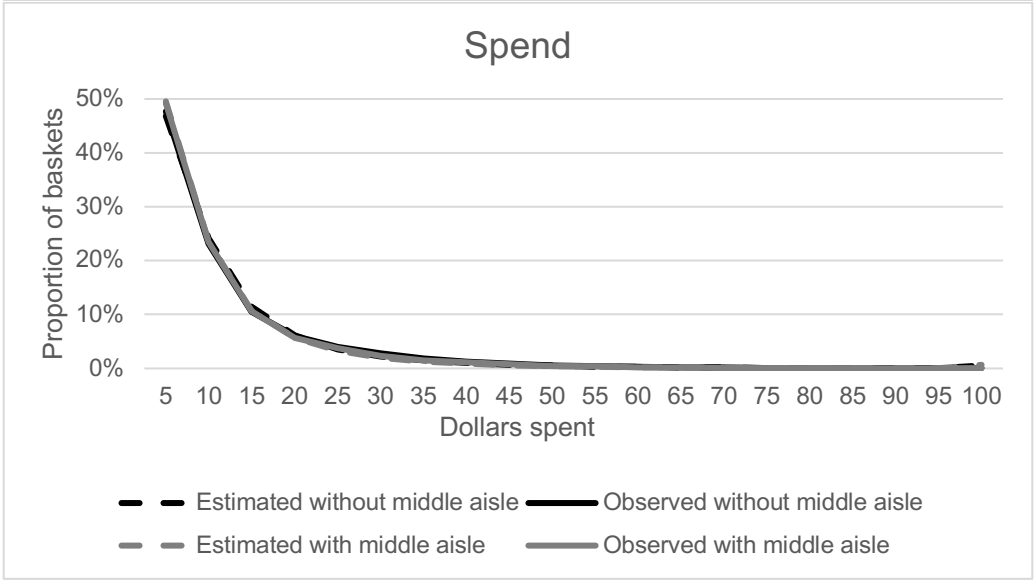
### RQ1: Basket size, spend, trip duration and efficiency

Figure 2 shows the comparison of the three observed distributions from the transaction data across the two stores. As we can see, the two stores are very close across all three distributions. Mann-Whitney U tests confirm that there is no statistically significant difference between the two stores by items (p=.88), dollar spend (p=.76), and duration of visit (p=.33). However, we do observe that the middle aisle store has a higher proportion of small basket size and quick trips than the standard store (see Table 1).

*Table 1: Proportion of short trips and small basket size across stores (transaction data)*

| Metric (%) | Without middle aisle | With middle aisle | P-value |
|---|---|---|---|
| Trip duration (<15 minutes) | 38 | 53 | <.01 |
| Items (<15 items) | 75 | 77 | <.01 |
| $ spend (<$15) | 81 | 83 | <.01 |



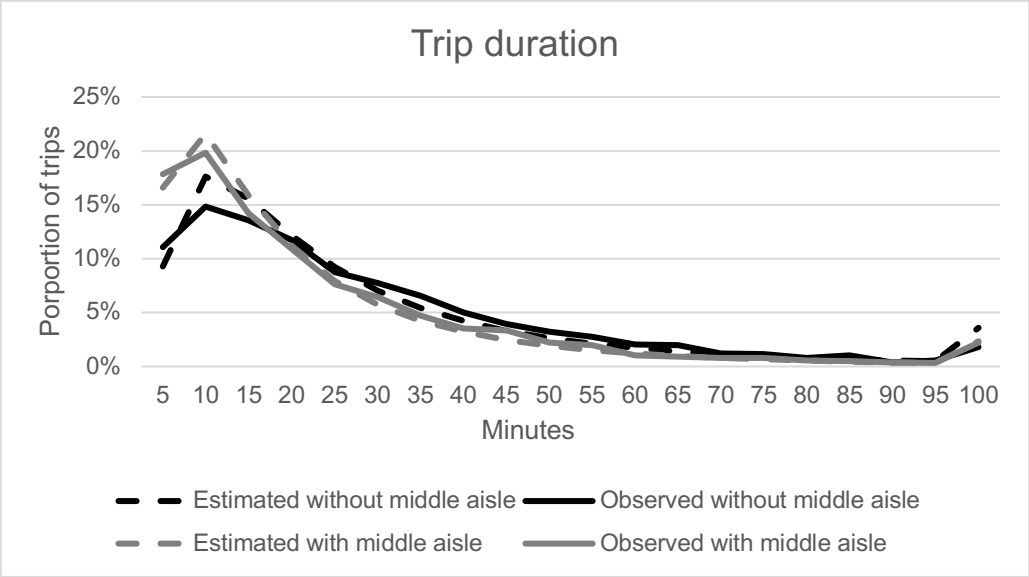



*Figure 2: Distributions of basket size, spend, and trip durations in the two stores. The store with a middle aisle has a higher number of quick trips than the standard store.*

The figures also show the fit of the Poisson lognormal model to items bought and the fit of the lognormal model to dollar spent and trip duration using transaction or Bluetooth data. The results show that both stores follow predictable patterns of items bought, dollar spend and trip duration despite having different retailing formats. There is no statistically significant difference between the observed distributions and the estimated distributions (see Table 2).

*Table 2: Mann-Whitney U tests of difference between estimated and observed transaction data*

| Metric/store | U value | P-value |
|---|---|---|
| Items/without middle aisle | 194.5 | .89 |
| Items/with middle aisle | 190 | .79 |
| Time visit/without middle aisle | 197 | .94 |
| Time visit/with middle aisle | 199 | .99 |
| $ spend/without middle aisle | 185 | .70 |
| $ spend/with middle aisle | 183 | .65 |

## RQ2: Shopper traffic in the middle aisle and endcap areas

For this exploratory question, researchers unobtrusively observed and recorded shopper traffic in terms of entering the middle aisle, exiting and interacting with the endcap displays in the middle aisle of Store 2.

### Aisle entry

As predicted by the recognized pattern of shopper behaviour known as the racetrack (Sorensen, 2016), shoppers were mostly at the back and sides of the whole store. Therefore, when observing the middle aisle in Store 2, two-thirds (64%) of shoppers approached the middle aisle from the back of the store through the normal aisles. No relationship was found between entering the

middle aisle from the front or back of the store and being in the first or second half $X^2$ (1, N = 777) = 1.01, p = .295.

*Table 2: Entry to middle aisle*

| %, n = 777 | Come from first half of store | Come from second half of store | Total |
|---|---|---|---|
| Come from back of store | 31 | 33 | 64 |
| Come from front of store | 16 | 20 | 36 |
| Total | 47 | 53 | 100 |

These observations suggest that the middle aisle endcaps facing the back of the store would have garnered twice as much exposure (potential to be in the shoppers' field of vision based on the direction of their movement) than the endcaps in the same middle aisle facing the front of the store.

### Aisle exit

The shopper traffic leaving the middle aisle was less clear, and more chaotic (See Figure 2). The most prevalent path (46% of shoppers) was to travel along the inner aisle, directly perpendicular to the middle aisle, continuing on as if it was not there. The exception to this was if shoppers were in the first half of the store, travelling from the back to the front (14% of all shoppers observed exiting the middle aisle, 46% of shoppers who entered the middle aisle in the first half of the store, from the back), where they were more likely to turn and return in the direction of the first half of the store (effectively doing a U-turn).

When travelling from the front to the back of the store, shoppers in the first half were still more likely to turn towards the front of the store than the back of the store, which suggests that shoppers used the middle aisle to return for forgotten items.

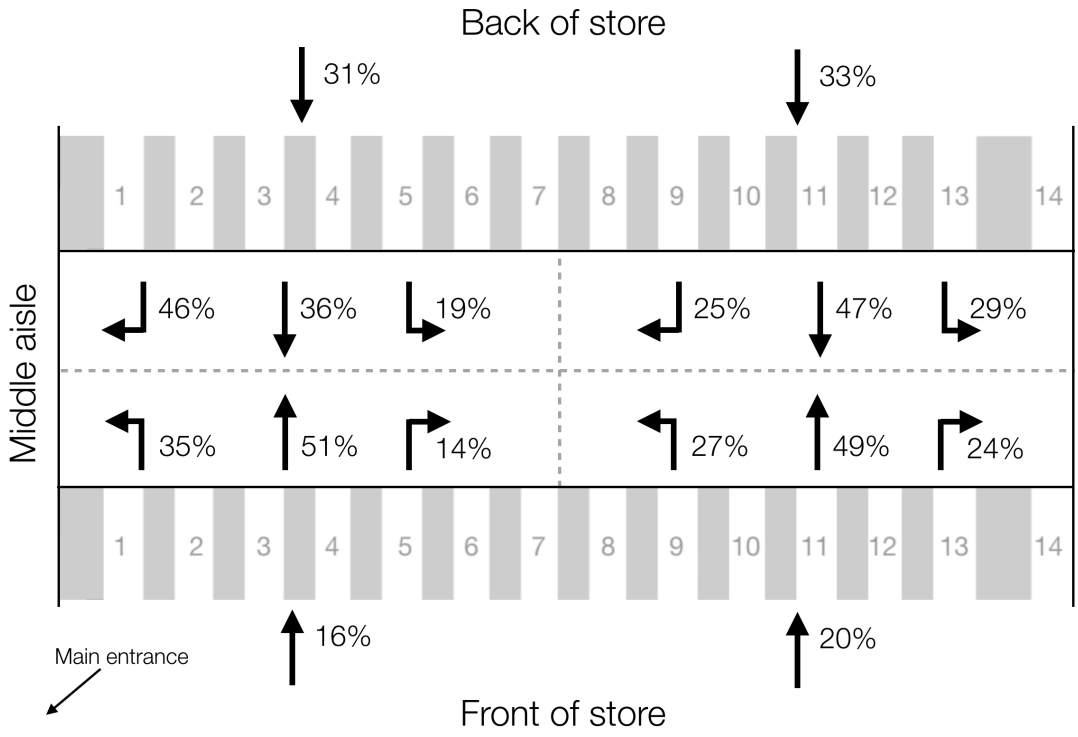



*Figure 3: Entry and exit to and from middle aisle*

*Table 3: Exit from middle aisle*

| Comes from: | n | Turns to first half (%) | Travels across (%) | Turns to second half (%) | Total |
|---|---|---|---|---|---|
| From first half of store, back | 239 | 46 | 36 | 19 | 100 |
| From second half of store, back | 256 | 25 | 47 | 29 | 100 |
| From first half of store, front | 128 | 35 | 51 | 14 | 100 |
| From second half of store, front | 154 | 27 | 49 | 24 | 100 |
| Average | | 33 | 46 | 21 | 100 |

## Endcap use

In a store with a middle aisle layout, the number of endcaps (promotional bays at the end of an aisle) is effectively double that amount in a conventional store design. Given that endcap spaces deliver the highest sales uplifts (Tan et al., 2018), and typically contain categories of impulse buying (e.g. chips, soft drinks, chocolate) (Caruso et al., 2017), it is conceivable to expect an increase in shopper interaction with the endcap display and purchasing from these bays.

The overall use of endcaps in the store with a middle aisle was lower than that in the standard layout. In the control store, with no middle aisle, 2.2% of all observed shoppers were interacting with an endcap (48% at rear, 52% at front of store), while in the store with the middle aisle, we saw only a total of 1.6%, (24% at the rear, 38% at the front, and 39% in the middle). The endcaps at the back and front of the middle aisle store were less used than the equivalent endcaps in the standard layout store. An explanation for this may be that shoppers may have an internal "promotional budget" which is spread across the entire store, and when they spend this additional budget in the mid-aisle endcaps, they may be more likely to disregard or ignore the ends of aisles more readily. However, we see slightly less (0.6% of all shoppers) use of endcaps in the middle aisle store, which may be related to the higher proportion of smaller trips in this store, or may simply be explainable by chance.

*Table 4: Endcap use in stores as a % of all shoppers observed*

| % | Mid aisle store | Standard store |
|---|---|---|
| Registers | 21 | 20 |
| All else | 78 | 77 |
| Endcaps | 1.56 | 2.22 |
| Back | 0.37 | 1.06 |
| Front | 0.59 | 1.16 |
| Mid, facing front | 0.27 | |
| Mid, facing back | 0.32 | |
| | 100 | 100 |

## Conclusions

**Implications for practitioners**

The central implication of this research is that the two layouts are very similar in measures of trip duration, basket size, and spend. The overall flow of shoppers through the middle aisle is from the back of the store to the front, consistent with the race track seen in standard stores. The promotional ends are, overall, used the same amount, which is effectively halving the use of each endcap. Unless the benefits to the retailer of selling more promotional ends outweigh the loss of overall shelf space, it does not appear to be worthwhile to lay out stores with a middle aisle.

**Implications for researchers**

Researchers who examine store performance across different layouts need to be caution when interpreting the data if they see any differences. As our study showed, the difference is mainly from the infrequently purchasing shoppers and the endcap areas. Overall, there is no significant difference between the two stores. Researchers therefore need to look at the disaggregate level data to understand changes in shopping behavior across store layouts in order to assess if these changes are significant and the underlying factors that cause these changes.

**Future research**

During the observation phase, customers in the middle aisle store reported frustration with the four-way intersections, and described the central intersections as crowded. As is the nature of field research, this was not predicted before the research took place, and so was not able to be examined. Consumers who perceive retail crowding are less likely to buy new brands, less likely to examine alternatives, and appear to be less likely to purchase at all (Lee, 1991). Increased crowding is related to lower satisfaction with shopping (Eroglu and Harrell Gilbert, 1986; Machleit et al., 2000; Pons et al., 2014), but is mediated by customer expectations – if consumers expect a store to be crowded (say, a supermarket on a Saturday morning), then they are more tolerant of it (Pons et

al., 2016). While actual crowding is important, crowding in stores exists on two dimensions: spatial and human (Machleit et al., 1994). It is the perception of crowding, and the perception of control, which mediates the effects of crowding (Lee, 1991). The perceptions of crowding may also have decreased consumer comfort, which is valued by consumers (Ainsworth and Foster, 2017).

Accordingly, future research regarding store layout should consider dimensions of crowding and comfort to more fully investigate the potential implications for shoppers of more complex layouts. Additionally, the advertising effect of the middle-of-store promotional space could be investigated by understanding which specific items are on the promotional ends, and determining if more is sold of these items, but perhaps from the aisle instead. That is, are shoppers seeing the ends in the middle aisle, but going to the primary location within the store to purchase.

That these two layouts are so similar in outcomes compels researchers and practitioners to understand alternative sources of influence. In-store mobile gamification has delivered mixed results; gamification decreases engagement with products, and appears to increase stress to the shopping trip, but if consumers *are* engaged with the game, then they are more likely to buy discounted products advertised within said game (Högberg et al., 2018). Price promotions, free gifts, and trials in-store have been found to have a great potential for driving sales of particular products (Nordfält and Lange, 2013; Roll and Pfeiffer, 2017), and the results of investigations seem to show that shoppers respond quickly and without much reflection, implying that changes to layout, adjacencies, and assortment are important operational decisions (Bauer et al., 2012; Nordfält and Lange, 2013).

**Limitations**

This is an exploratory, observational study of a single example of a store with an alternative layout. As such, it has little generalizability, but remains an important first step in documenting differences in store layouts, and the potential for improving such layouts. It builds and extends knowledge of patterns of store layouts; that is the ubiquity of the racetrack and the flow around the store. The use of mobile phones while shopping may have an effect on shopping behaviour but was not observed in this study. This study is also narrowly focused on outcomes of interest to the retailer; customer and supplier perspectives have not been taken into consideration.

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**References**


Ainsworth, J., Foster, J., 2017. Comfort in brick and mortar shopping experiences: Examining antecedents and consequences of comfortable retail experiences. Journal of Retailing and Consumer Services 35, 27-35.

Altuntas, S., 2017. A novel approach based on utility mining for store layout: a case study in a supermarket. Industrial Management & Data Systems 117 (2), 304-319.

Anic, I.-D., Radas, S., Lim, L.K.S., 2010. Relative effects of store traffic and customer traffic flow on shopper spending. The International Review of Retail, Distribution and Consumer Research 20 (2), 237-250.

Bauer, J.C., Kotouc, A.J., Rudolph, T., 2012. What constitutes a "good assortment"? A scale for measuring consumers' perceptions of an assortment offered in a grocery category. Journal of Retailing and Consumer Services 19 (1), 11-26.

Bensinger, G., Morris, K., 2014. Amazon to open first brick-and-mortar site. Wall Street Journal, New York City.

Bezawada, R., Balachander, S., Kannan, P.K., Shankar, V., 2009. Cross-category effects of aisle and display placements: A spatial modeling approach and insights. Journal of Marketing 73 (3), 99-117.

Bogomolova, S., Szabo, M., Kennedy, R., 2017. Retailers' and manufacturers' price-promotion decisions: Intuitive or evidence-based? Journal of Business Research 76 (1), 189-200.

Caruso, W., Bogolovema, S., Corsi, A., Maria,, Cohen, J., Lockshin, L., Tan, P., 2017. The Real Estate Value of Supermarket Endcaps: Why Location In-Store Matters. Journal of Advertising Research 58 (2), 177-188.

Chevalier, M., 1975. Increase in sales due to in-store display. Journal of Marketing Research 12 (November), 426-431.

Desmet, P., Renaudin, V., 1998. Estimation of product category sales responsiveness to allocated shelf space. International Journal of Research in Marketing 15 (5), 443-457.

Ehrenberg, A., 1969. Towards an integrated theory of consumer behaviour. Journal of the Market Research Society 11 (4), 305-337.

Ehrenberg, A., 1995. Empirical generalisations, theory, and method. Marketing Science 14 (3), G20-G28.

Eroglu, S., Harrell Gilbert, 1986. Retail Crowding: Theoretical and Strategic Implications. Journal of Retailing 62 (4), 346-363.

Farley, J., Ring, W., 1966. A Stochastic Model of Supermarket Traffic Flow. Operations Research 14 (4), 555-567.

Hagberg, J., Sundstrom, M., Egels-Zandén, N., 2016. The digitalization of retailing: an exploratory framework. International Journal of Retail & Distribution Management 44 (7), 694-712.

Högberg, J., Shams, P., Wästlund, E., 2018. Gamified in-store mobile marketing: The mixed effect of gamified point-of-purchase advertising. Journal of Retailing and Consumer Services.

Hui, S.K., Inman, J.J., Huang, Y., Suher, J.A., 2012. Estimating the Effect of In-Store Travel Distance on Unplanned Spending: Applications to Store Layout and Mobile Promotion Strategies. Citeseer.

Juel-Jacobsen, L.G., 2015a. Aisles of life: outline of a customer-centric approach to retail space management. The International Review of Retail, Distribution and Consumer Research 25 (2), 162-180.

Juel-Jacobsen, L.G., 2015b. "The World for a Crooked Street": Towards a Supermarket Morphology of Shopping Aisles and Retail Layout. Interiors 6 (1), 59-89.
Larson, R., 2006. Core principles for supermarket aisle management. Journal of Food Distribution Research 37 (1), 101-105.
Lee, W., 1991. Consumer in-store exploratory behavior: the effects of perceived retail crowding and cognitive control, Business Administration. Pennsylvania State University, United States, pp. 1-192.
Licoppe, C., 2013. Merging mobile communication studies and urban research: Mobile locative media,"onscreen encounters" and the reshaping of the interaction order in public places. Mobile Media & Communication 1 (1), 122-128.
Machleit, K.A., Eroglu, S.A., Mantel, S.P., 2000. Perceived retail crowding and shopping satisfaction: what modifies this relationship? Journal of consumer psychology 9 (1), 29-42.
Machleit, K.A., Kellaris, J.J., Eroglu, S.A., 1994. Human versus spatial dimensions of crowding perceptions in retail environments: a note on their measurement and effect on shopper satisfaction. Marketing Letters 5 (2), 183-194.
Mowrey, C.H., Parikh, P.J., Gue, K.R., 2017. The impact of rack layout on visual experience in a retail store. INFOR: Information Systems and Operational Research, 1-24.
Nakamura, R., Pechey, R., Suhrcke, M., Jebb, S.A., Marteau, T.M., 2014. Sales impact of displaying alcoholic and non-alcoholic beverages in end-of-aisle locations: An observational study. Social Science & Medicine 108 (1), 68-73.
Nordfält, J., Lange, F., 2013. In-store demonstrations as a promotion tool. Journal of Retailing and Consumer Services 20 (1), 20-25.
Page B, Sharp A, Lockshin L, H, S., 2017. Parents and Children in Supermarkets: Incidence and Influence. Journal of Retailing and Consumer Services.
Parikh, P.J., Mowrey, C., 2014. Designing Retail Facilities to Improve Product Exposure, 13th IMHRC, Cincinnati, Ohio, pp. 1-17.
Phua, P., Page, B., Bogomolova, S., 2015. Validating Bluetooth logging as metric for shopper behaviour studies. Journal of Retailing and Consumer Services 22, 158-163.
Pizzi, G., Scarpi, D., 2016. The effect of shelf layout on satisfaction and perceived assortment size: An empirical assessment. Journal of Retailing and Consumer Services 28, 67-77.
Pons, F., Giroux, M., Mourali, M., Zins, M., 2016. The relationship between density perceptions and satisfaction in the retail setting: Mediation and moderation effects. Journal of Business Research 69 (2), 1000-1007.
Pons, F., Mourali, M., Giroux, M., 2014. The density–satisfaction relationship revisited: The role of scarcity and consumers affective reactions in a crowded retail situation. Journal of Retailing and Consumer Services 21 (1), 54-60.
Roll, O., Pfeiffer, E., 2017. Are your consumers variety seekers? The moderating impact on the effectiveness of free gift promotions vs. price discounts. The International Review of Retail, Distribution and Consumer Research 27 (4), 352-368.
Scamell-Katz, S., 2012. The Art of Shopping: How We Shop and Why We Buy. LID Publishing, UK.

Sharp, B., 2010. How brands grow, in: Sharp, B. (Ed.), How Brands Grow. Oxford University Press, South Melbourne, Australia, pp. 16-27.
Smith, A., Anderson, M., 2016. Online Shopping and E-Commerce. Pew Research Center, Washington, D.C, pp. 1-30.
Smith, P., Burns, D., 1996. Atmospherics and retail environments: the case of the "power aisle". International Journal of Retail & Distribution Management 24 (1), 7.
Sommer, R., Aitkens, S., 1982. Mental mapping of two supermarkets. Journal of Consumer Research, 211-215.
Sorensen, H., 2009. Inside the mind of the shopper. Pearson Education Inc., Upper Saddle River, New Jersey.
Sorensen, H., 2016. Inside the mind of the shopper: The science of retailing, 2 ed. Pearson Education, New Jersey, United States.
Sorensen, H., Bogomolova, S., Anderson, K., Trinh, G., Sharp, A., Kennedy, R., Page, B., Wright, M., 2017. Fundamental patterns of in-store shopper behavior. Journal of Retailing and Consumer Services 37, 182-194.
Tan, P.J., Corsi, A., Cohen, J., Sharp, A., Lockshin, L., Caruso, W., Bogomolova, S., 2018. Assessing the sales effectiveness of differently located endcaps in a supermarket. Journal of Retailing and Consumer Services 43, 200-208.
U.S. Census Bureau News, 2018. Quarterly Retail E-Commerce Sales 2nd Quarter 2018. U.S. Department of Commerce, Washington, D.C., pp. 1-3.
Underhill, P., 1999. Why We Buy: The Science of Shopping. Obat, Inc, London.
Wilkinson, J.B., Mason, J.B., Paksoy, C.H., 1982. Assessing the impact of short-term supermarket strategy variables. Journal of Marketing Research 19 (February), 72-86.
Yapicioglu, H., Smith, A.E., 2012. Retail space design considering revenue and adjacencies using a racetrack aisle network. IIE Transactions 44 (6), 446-458.